CAPSTONE PROJECT

DETAILED ANSWERS TO QUESTIONS

DATA SET USED: Vehicle Insurance Claim Fraud Detection

STEP 1 - Deliverable: Clear problem statement + task type + target metric.

1. Frame the business and data science problem clearly.
2. Define whether it's a classification, regression, recommendation, anomaly detection, or clustering task.
3. Specify success metrics (e.g., Accuracy, AUC, RMSE, Silhouette Score) and business KPIs (e.g., cost savings, uplift).
4. Capstone linkage: Module 1 output maps to Capstone Steps 1–3.

ANSWERS

1. This project aims to develop a machine learning-based fraud detection model for car insurance claims using the Fraud Oracle dataset. Insurance fraud creates significant financial and operational challenges by increasing claim losses and requiring costly manual investigations. To address this, the project will use historical claims data to predict whether a claim is likely to be fraudulent.

2. The project is formulated as a ***binary classification problem***, where the target variable is **FraudFound_P**, representing whether fraud was detected in the claim. The objective is to build a predictive model that can support insurers in automatically flagging suspicious claims for further review, thereby improving fraud detection efficiency, reducing fraudulent payouts, and optimizing investigative resources.

3. The model will be evaluated primarily using **Recall**, as the key business priority is to identify as many fraudulent claims as possible and minimize missed fraud cases. In fraud detection, failing to detect fraudulent claims can lead to direct financial loss, making Recall the most business-relevant primary metric.

   As a secondary metric, **ROC-AUC** will be used to assess the model's ability to distinguish between fraudulent and legitimate claims across different classification thresholds. Additional supporting metrics such as **Precision**, **F1-score**, and **Accuracy** will also be reported to provide a balanced view of predictive performance and operational trade-offs.

Metric Hierarchy

1. **Primary Target Metric:** Recall
2. **Secondary Target Metric:** ROC-AUC
3. **Supporting Metrics:** Precision, F1-score, Accuracy

4. The business success of the fraud detection model can be evaluated using operational and financial key performance indicators. Proposals are as follows:
   a. **Fraud Detection Rate** will measure the proportion of fraudulent claims successfully identified for further investigation.
   b. **False positive rate** will also be monitored to avoid excessive flagging of legitimate claims. Overall, the model is expected to improve fraud detection while balancing cost savings, operational efficiency, and customer experience.
   c. **Reduction in fraud-related claim losses** by preventing payouts on suspicious or fraudulent claims.

d. **Review Rate** will be assessed by measuring the reduction in the number of claims requiring manual review.

This capstone addresses a supervised machine learning problem in the insurance domain: predicting whether a car insurance claim is fraudulent using historical claims data. The business objective is not to automate final claim denial, but to support earlier fraud screening and smarter investigation prioritization.

STEP 2 – Data Collection & Understanding

1. Summarise feature types, missing values, outliers, etc.
2. Provide a data dictionary (variables, types, units, allowed values).

- **DATA USED: LINK:** Vehicle Insurance Claim Fraud Detection
- **DATA SUMMARY:** The dataset used for this project is fraud_oracle.csv, which contains **15,420 automobile insurance claim records** and **33 variables**. Each observation represents a single insurance claim and includes information about the policyholder, vehicle, claim timing, claim circumstances, and fraud outcome. The target variable is **FraudFound_P**, where 0 indicates a non-fraudulent claim and 1 indicates a fraudulent claim. This dataset is suitable for the prescribed fraud detection domain because it provides labeled historical claims data that can support supervised machine learning.

  The dataset contains a mixture of **categorical/ordinal variables** and **numeric/discrete variables**. Categorical features include claim timing categories, vehicle make, accident area, policy type, vehicle price bands, prior claims history, and investigation-related indicators such as police report filing and witness presence. Numeric features include age, deductible, driver rating, week-of-month indicators, representative number, and year. Several variables, such as VehiclePrice, PastNumberOfClaims, AgeOfVehicle, and NumberOfCars, are categorical but inherently ordinal because they represent ordered ranges or grouped bands.

  A review of data quality shows that the dataset contains **no formal missing values**, as all variables have complete entries. However, several columns contain **special placeholder values** that may require preprocessing. For example, DayOfWeekClaimed and MonthClaimed include the value "0", which likely represents an unknown or invalid category, while Days_Policy_Accident and Days_Policy_Claim include the value "none", which may represent unavailable or non-applicable timing information. In addition, the Age variable contains a minimum value of **0**, which is not realistic for a policyholder and should be treated as a potential anomaly or data quality issue. The PolicyNumber variable is a unique identifier and should be excluded from predictive modeling.

  The target variable is **imbalanced**, with **14,497 non-fraud cases** and **923 fraud cases**, meaning only about **6% of claims are fraudulent**. This is consistent with real-world fraud detection problems and reinforces the need to use performance metrics beyond simple accuracy. Overall, the dataset is structurally strong for the project, but it requires careful preprocessing of placeholder values, potential anomalies, and categorical encoding before model development.

  Why this dataset is appropriate:

  The Fraud Oracle dataset is suitable for this capstone because it contains labeled historical insurance claims and includes variables related to policyholder profile, policy structure, accident circumstances, and claim handling. This makes it appropriate for supervised fraud detection, where the goal is to learn patterns associated with fraudulent versus legitimate claims.

The dataset also reflects a realistic fraud challenge because the fraud class is highly imbalanced, which mirrors real-world insurance environments where fraud cases are relatively rare.

Data Dictionary

NO	Variable	Description	Data Type	Unit / Scale	Allowed Values / Examples	Notes
1	**Month**	Month when the accident occurred	Categorical (Nominal)	Calendar month	Jan, Feb, Mar, …, Dec	Incident month
2	**WeekOfMonth**	Week number of the month when the accident occurred	Numeric (Discrete)	Integer	1–5	Incident timing
3	**DayOfWeek**	Day of the week when the accident occurred	Categorical (Nominal)	Day name	Monday–Sunday	Incident day
4	**Make**	Vehicle manufacturer / brand	Categorical (Nominal)	Brand label	Honda, Toyota, Ford, BMW, Ferrari, etc.	Vehicle make
5	**AccidentArea**	Location type of accident	Categorical (Binary)	Area type	Urban, Rural	Claim context
6	**DayOfWeekClaimed**	Day of the week when claim was filed	Categorical (Nominal)	Day name	Monday–Sunday, 0	0 likely placeholder/unknown
7	**MonthClaimed**	Month when claim was filed	Categorical (Nominal)	Calendar month	Jan–Dec, 0	0 likely placeholder/unknown
8	**WeekOfMonthClaimed**	Week number of month when claim was filed	Numeric (Discrete)	Integer	1–5	Claim timing
9	**Sex**	Policyholder gender	Categorical (Binary)	Label	Male, Female	Sensitive attribute (for fairness checks)
10	**MaritalStatus**	Marital status of policyholder	Categorical (Nominal)	Label	Single, Married, Divorced, Widow	Demographic feature
11	**Age**	Age of policyholder	Numeric (Discrete)	Years	0–80 observed	0 is likely invalid/anomalous
12	**Fault**	Party at fault in accident	Categorical (Binary)	Label	Policy Holder, Third Party	Claim liability indicator
13	**PolicyType**	Detailed policy type	Categorical (Nominal)	Policy label	Sedan/Sport/Utility × Liability/Collision/All Perils	Composite policy variable
14	**VehicleCategory**	Vehicle class	Categorical (Nominal)	Category	Sedan, Sport, Utility	Vehicle segment
15	**VehiclePrice**	Vehicle price band	Categorical (Ordinal)	Price range band	less than 20000; 20000 to 29000; 30000 to 39000; 40000 to 59000; 60000 to 69000; more than 69000	Ordinal band
16	**FraudFound_P**	Fraud label (target variable)	Numeric (Binary)	0/1	0 = No fraud, 1 = Fraud	Target variable
17	**PolicyNumber**	Unique policy identifier	Numeric (Identifier)	Integer ID	1–15420	Should be treated as ID, not predictive feature
18	**RepNumber**	Representative / agent code	Numeric (Discrete Code)	Integer code	1–16	Likely categorical code
19	**Deductible**	Deductible amount under policy	Numeric (Discrete)	Currency amount	300, 400, 500, 700	Monetary feature

20	**DriverRating**	Driver risk / rating score	Numeric (Ordinal)	Rating scale	1–4	Lower/upper risk depends on source context
21	**Days_Policy_Accident**	Time between policy start and accident	Categorical (Ordinal)	Time range	none, 1 to 7, 8 to 15, 15 to 30, more than 30	Ordered band
22	**Days_Policy_Claim**	Time between policy start and claim filing	Categorical (Ordinal)	Time range	none, 8 to 15, 15 to 30, more than 30	Ordered band
23	**PastNumberOfClaims**	Number of prior claims	Categorical (Ordinal)	Count band	none, 1, 2 to 4, more than 4	Historical claims feature
24	**AgeOfVehicle**	Age of vehicle	Categorical (Ordinal)	Age band	new, 2 years, 3 years, 4 years, 5 years, 6 years, 7 years, more than 7	Ordered band
25	**AgeOfPolicyHolder**	Age band of policyholder	Categorical (Ordinal)	Age range	16 to 17, 18 to 20, 21 to 25, 26 to 30, 31 to 35, 36 to 40, 41 to 50, 51 to 65, over 65	Age grouping
26	**PoliceReportFiled**	Whether police report was filed	Categorical (Binary)	Yes/No	Yes, No	Important fraud signal candidate
27	**WitnessPresent**	Whether a witness was present	Categorical (Binary)	Yes/No	Yes, No	Important fraud signal candidate
28	**AgentType**	Type of agent handling policy	Categorical (Binary)	Label	Internal, External	Distribution channel feature
29	**NumberOfSupplements**	Number of supporting documents / supplements	Categorical (Ordinal)	Count band	none, 1 to 2, 3 to 5, more than 5	Ordered band
30	**AddressChange_Claim**	Time since address change before claim	Categorical (Ordinal)	Time band	under 6 months, 1 year, 2 to 3 years, 4 to 8 years, no change	Ordered band
31	**NumberOfCars**	Number of vehicles associated with policyholder	Categorical (Ordinal)	Count band	1 vehicle, 2 vehicles, 3 to 4, 5 to 8, more than 8	Household / exposure feature
32	**Year**	Year of record / incident	Numeric (Discrete)	Calendar year	1994, 1995, 1996	Time context
33	**BasePolicy**	Base coverage type	Categorical (Nominal)	Policy label	Liability, Collision, All Perils	Simplified policy class

STEP 3

Data Cleansing - The dataset was reviewed for invalid, placeholder, and anomalous values before preprocessing.

- One invalid category was found in DayOfWeekClaimed, where "0" appeared instead of a valid weekday. This was recoded to "Unknown".
- One invalid category was also found in MonthClaimed, where "0" appeared instead of a valid month. This was also recoded to "Unknown".
- The value "none" appeared in Days_Policy_Accident and Days_Policy_Claim. These were retained as valid categories because they may represent unavailable or non-applicable timing information rather than true missing data.
- The variable Age contained 320 records with a value of 0, which is not realistic for a policyholder. These were treated as invalid values.
- The invalid age values were converted to missing and then imputed using the midpoint of the corresponding AgeOfPolicyHolder band.
- Median imputation was used only as a fallback in case any remaining age values were still missing.

- PolicyNumber was removed from the modelling dataset because it is a unique identifier and not a meaningful predictive feature.

Feature Engineering

Feature engineering was applied. Categorical variables were encoded to make them usable for machine learning models, while numeric variables were scaled to support algorithms that are sensitive to feature magnitude and to prepare the dataset for PCA. Domain-derived features were also created to capture potentially meaningful fraud signals, such as whether a police report was filed, whether a witness was present, whether the vehicle belonged to a high price band, and whether the policyholder had a recent address change. These engineered features were added to improve model interpretability and potentially strengthen fraud prediction.

EDA Applied

Exploratory Data Analysis (EDA) was conducted to understand the structure of the dataset and identify patterns related to fraud. The target variable FraudFound_P is highly imbalanced, with **14,497 non-fraud cases** and **923 fraud cases**, meaning only about **5.99%** of records are fraudulent. This confirms that the dataset is a class-imbalanced classification problem, so model evaluation should focus on Recall, Precision, F1-score, and ROC-AUC rather than Accuracy alone.

Fraud rates were then examined across selected categorical variables. Claims where the **policy holder** was at fault had a much higher fraud rate than claims attributed to a third party. Fraud rates were also noticeably different across BasePolicy, VehicleCategory, and AccidentArea. For example, **All Perils** policies showed a higher fraud rate than **Liability** policies, and **Rural** accidents showed a slightly higher fraud rate than **Urban** accidents. These findings suggest that fraud is not randomly distributed and that several business-related variables carry predictive signal.

Because this project is a supervised fraud classification problem rather than an unsupervised segmentation task, clustering tendency analysis is not a core requirement for the modelling objective. Therefore, the EDA focused on class distribution, feature distributions, and fraud relationships rather than cluster tendency.

EDA Highlights and Modelling Implications:
The exploratory analysis confirmed that the dataset is highly imbalanced, with fraud representing only a small portion of total claims. This reinforced the decision to prioritise Recall, F1 Score, and ROC-AUC over Accuracy when comparing models.

EDA also showed that fraud is not randomly distributed across the dataset. Variables such as **Fault**, **BasePolicy**, **VehicleCategory**, and **AccidentArea** showed noticeably different fraud patterns. This supports the use of tree-based models, which are better able to capture non-linear interactions and complex patterns in structured tabular data.

In addition, the data-cleaning stage identified invalid placeholders and anomalous values, especially in date-related claim fields and the **Age** variable. These findings directly informed the preprocessing strategy, feature engineering decisions, and model selection process.

Feature importance and explainability

To explain which variables were most influential in fraud prediction, a **model-based feature importance** approach was used. A Random Forest classifier was trained after preprocessing, and feature importance values were extracted from the fitted model. This method is suitable for the project because it is practical, interpretable, and easy to justify in a business-facing capstone.

The most influential features were expected to include accident responsibility, policy type, base policy, vehicle category, age, and selected claim timing variables. This indicates that fraud detection depends on a combination of claim context, policy structure, and customer profile characteristics. Model-based importance was chosen as the main explainability approach because it is simpler to implement and defend than SHAP or LIME at this stage, while still meeting the requirement for explainability.

Feature selection

A **filter-based feature selection** method was applied using **Mutual Information**. This method measures how strongly each feature is associated with the target variable and helps identify the most informative variables before modelling. Mutual Information was selected because it is straightforward, works well with classification tasks, and is easy to explain in the report.

This step helped confirm that not all transformed features contribute equally to fraud prediction. It also supports a more disciplined modelling process by highlighting variables with stronger predictive relevance.

Dimensionality reduction

After encoding the categorical variables, the feature space became much wider. To address this, **Principal Component Analysis (PCA)** was applied after scaling and transformation. PCA was used to examine how many principal components are needed to retain most of the variance in the dataset.

This step helps reduce dimensionality, simplify the feature space, and support experimentation with more compact model inputs. PCA was mainly used here as an exploratory dimensionality reduction technique rather than as the only modelling input. Since the project is classification-focused, PCA complements the analysis but does not replace the original engineered features.

Overall, Step 3 established a defensible modelling foundation by combining data cleaning, applied EDA, domain-informed feature engineering, feature selection using Mutual Information, and dimensionality

reduction using PCA. This ensured that subsequent model comparisons were performed on a dataset that was both analytically structured and operationally meaningful.

SUMMARY OF ACTIONS

- The dataset was cleaned first before feature engineering was performed.
- DayOfWeekClaimed = "0" was changed to "Unknown" because "0" is not a valid day of the week.
- MonthClaimed = "0" was changed to "Unknown" because "0" is not a valid month.
- The value "none" in Days_Policy_Accident and Days_Policy_Claim was retained as its own category because it may represent unavailable or non-applicable information.
- Age = 0 was treated as an invalid value because a policyholder cannot realistically have an age of zero.
- Invalid age values were replaced using the midpoint of the corresponding AgeOfPolicyHolder age band.
- PolicyNumber was removed because it functions only as an identifier and does not contribute predictive value for fraud detection.
- Categorical variables were encoded so that they could be processed by machine learning models.
- Numeric variables were scaled to prepare them for PCA and modelling.
- Additional features were created, such as police report flag, witness flag, high vehicle price flag, and recent address change flag.
- The fraud target distribution was examined, and the dataset was confirmed to be imbalanced.
- Variables that appeared more related to fraud, such as fault, base policy, vehicle category, and accident area, were explored.
- Random Forest feature importance was used as the explainability method.
- Mutual Information was used as the feature selection method.
- PCA was applied to reduce the dimensionality of the transformed dataset and to assess how much variance could be retained using fewer components.
- Since the project is a supervised classification problem, clustering tendency was not treated as the main focus of the EDA.

STEP 4 - Model Implementation
INSTRUCTIONS:

- Experiment with appropriate models:
    - **Supervised**: Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, etc.
    - **Unsupervised**: K-Means, DBSCAN, Hierarchical (Elbow, Silhouette).
    - **Recommendation**: Collaborative or content-based.
    - **Deep Learning**: RNNs, CNNs, LSTMs, Transformers (if appropriate).
- Evaluation: compare with relevant metrics.
- **Reproducibility**: save configs and artefacts (models/).

Deliverables: Trained models, metrics and comparison between models.

STEPS PERFORMED

- Since the selected dataset contains a labeled fraud target (FraudFound_P), supervised learning was identified as the most appropriate modelling approach.
- Four supervised classification models were selected for experimentation: Logistic Regression, Decision Tree, Random Forest, and XGBoost (or Gradient Boosting as a fallback).
- Logistic Regression was included as a simple and interpretable baseline model.
- Decision Tree was included because it is easy to interpret and can capture nonlinear rules.

- Random Forest was included because it is a strong ensemble model that performs well on structured tabular data.
- XGBoost (or Gradient Boosting) was included because boosting methods are often highly effective for fraud detection problems.
- Unsupervised models such as K-Means, DBSCAN, and Hierarchical Clustering were not prioritized because the dataset already contains a labeled fraud target, making supervised classification more suitable.
- Recommendation system methods were not applicable because the project does not involve user-item preference prediction.
- Deep learning methods were also not prioritized because the dataset is structured tabular data and does not require image, sequence, or text modelling.
- Model performance was compared using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- Since the dataset is imbalanced, Recall, F1-score, and ROC-AUC were treated as more important than Accuracy when selecting the best model.
- All trained models were saved as reproducible artefacts in a models/ folder for future reuse and validation.

To address the business problem of identifying potentially fraudulent insurance claims, multiple supervised machine learning models were developed and compared. Since the target variable **FraudFound_P** is binary (fraud vs. non-fraud), the problem was framed as a **supervised classification task**. The objective was not simply to maximise overall accuracy, but to find a model that could meaningfully detect fraudulent claims while balancing the trade-offs between precision, recall, and operational usability.

Model Selection Approach

Several classification models were tested to evaluate their suitability for fraud detection. The following supervised learning models were included in the experiment:

- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **Gradient Boosting**

These models were selected because they represent a useful mix of:

- **baseline linear classification** (Logistic Regression),
- **interpretable tree-based learning** (Decision Tree),
- **ensemble bagging methods** (Random Forest), and
- **ensemble boosting methods** (Gradient Boosting).

This provided a balanced comparison between simpler and more advanced classifiers.

Data Preparation for Modelling

Before model training, the dataset underwent preprocessing and feature engineering to ensure that the models could learn meaningful patterns from the insurance claims data.

The following preparation steps were applied:

- Invalid placeholder values such as "0" in **DayOfWeekClaimed** and **MonthClaimed** were recoded to "Unknown".

- Invalid age values (Age = 0) were treated as missing and imputed using the midpoint of the corresponding **AgeOfPolicyHolder** band.
- **PolicyNumber** was removed from the modelling dataset because it functions as an identifier rather than a predictive feature.
- Additional engineered features were created to improve signal quality, including:
    - **HasPoliceReport**
    - **HasWitness**
    - **HighVehiclePrice**
    - **RecentAddressChange**
    - **ShortPolicyToAccident**

Because the dataset contains both numerical and categorical variables, a preprocessing pipeline was used:

- **Numerical features** were imputed using the median and scaled using **StandardScaler**
- **Categorical features** were imputed using the most frequent value and encoded using **OneHotEncoder**

A **train-test split of 80:20** was used, with **stratification on the fraud target** to preserve the class distribution in both sets.

Model Evaluation Metrics

Because insurance fraud detection is an **imbalanced classification problem**, multiple evaluation metrics were used instead of relying on accuracy alone. The following metrics were used to compare models:

- **Accuracy** – overall percentage of correct predictions
- **Precision** – proportion of predicted fraud cases that were actually fraud
- **Recall** – proportion of actual fraud cases correctly detected
- **F1-Score** – balance between precision and recall
- **ROC-AUC** – ability to distinguish between fraud and non-fraud across thresholds

In fraud detection, **recall** is especially important because missing a fraudulent claim can lead to direct financial loss. However, precision is also operationally relevant because excessive false positives can burden fraud investigation teams.

Model Comparison Results

The following results were obtained from the model comparison:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.643320	0.131939	0.886486	0.229692	0.806250
Decision Tree	0.632944	0.127459	0.875676	0.222527	0.798809
Random Forest	0.676719	0.136652	0.827027	0.234843	0.815103
Gradient Boosting	0.940661	0.666667	0.021622	0.041885	0.840869

Interpretation of Model Results

The results show that **Gradient Boosting** achieved the highest **accuracy (94.07%)** and the highest **ROC-AUC (0.8409)**. At first glance, this may appear to be the best model. However, a closer examination reveals that its

recall is only 2.16%, meaning it failed to identify the overwhelming majority of actual fraud cases. This is a critical weakness in a fraud detection context. The model is likely over-optimised toward the majority non-fraud class and therefore unsuitable as the primary fraud-screening model.

Logistic Regression and **Decision Tree** both produced high recall (88.65% and 87.57%, respectively), which means they were effective at detecting most fraud cases. However, their precision was low (around 12–13%), indicating a large number of false positives. This would create significant operational inefficiency if used directly in a real fraud investigation workflow.

Among the tested models, **Random Forest** offered the most balanced performance. It achieved:

- **Accuracy = 67.67%**
- **Precision = 13.67%**
- **Recall = 82.70%**
- **F1-Score = 0.2348**
- **ROC-AUC = 0.8151**

Although its accuracy is lower than Gradient Boosting, it retained a **strong fraud detection rate** while providing the **best F1-Score among the high-recall models**. This indicates a more practical balance between detecting fraud and controlling the volume of false alarms.

Selected Final Model

Based on the model comparison, **Random Forest** was selected as the **final candidate model** for this fraud detection use case.

The selection was based on the following rationale:

- It maintained **high recall**, which is essential for catching fraudulent claims.
- It achieved the **best balance** between precision and recall among the practically usable models.
- It produced a competitive **ROC-AUC**, indicating good ranking ability.
- It is more robust than a single decision tree and less operationally misleading than a high-accuracy/low-recall model like Gradient Boosting.
- It supports **feature importance analysis**, which is valuable for interpretability and governance.

From a business perspective, the selected model is best suited for **fraud investigation prioritisation**, where flagged claims are reviewed by investigators rather than automatically rejected.

Operational Role of the Final Model:

The Original Random Forest was selected as the final predictive tool because it provided the best practical balance between fraud capture, ranking ability, and investigation usability. In deployment terms, it is most appropriate as a fraud-screening and triage model, where flagged claims are routed to human investigators rather than automatically rejected.

Why the highest-accuracy model was not selected:
Although Gradient Boosting achieved the highest Accuracy and ROC-AUC, it was not selected because its Recall was extremely low. In a fraud detection context, this is a major weakness because a model that misses most fraudulent claims is not operationally useful, even if it looks strong on Accuracy.

This comparison confirms that the most appropriate predictive model is not the one with the most impressive surface-level metric, but the one that best aligns with the business objective of identifying suspicious claims for further review.

Reproducibility and Artefacts

To support reproducibility, the modelling workflow was structured using consistent preprocessing pipelines and saved outputs. The model comparison metrics were exported to:

- **models/model_comparison_metrics.csv**

This ensures that model results can be reviewed, audited, and reproduced consistently. In a production-ready version of the project, additional artefacts such as the trained model object, preprocessing pipeline, and configuration files should also be stored in a dedicated **models/** directory.

Step 4 Conclusion

The model comparison confirms that the highest-accuracy model is not necessarily the best business model in an imbalanced fraud detection problem. While Gradient Boosting appeared strongest on surface-level metrics, its extremely poor recall made it unsuitable for detecting fraud. In contrast, **Random Forest** delivered the most practical and defensible balance between fraud detection capability and operational feasibility. Therefore, it was selected as the preferred model for the next stage of ethical review and fairness auditing.

STEP 5 - Bias & Fairness Analysis (Ethical AI & Critical Thinking)

To validate the fraud detection approach, multiple classification models were evaluated using performance metrics appropriate for imbalanced fraud data, including **Accuracy, Precision, Recall, F1 Score, and ROC-AUC**. Because fraudulent claims are relatively rare, the evaluation focused more heavily on **Recall, F1 Score, and ROC-AUC**, rather than Accuracy alone.

The **Original Random Forest** emerged as the **best overall predictive model** among the practically viable options. It achieved an **Accuracy of 0.6767**, **Precision of 0.1367**, **Recall of 0.8270**, **F1 Score of 0.2348**, and **ROC-AUC of 0.8151**. These results show that the model was able to capture a large proportion of fraudulent claims while maintaining the strongest overall balance between fraud detection and false-positive control. In fraud detection, this is especially important because the objective is not simply to be correct most of the time, but to identify suspicious claims early enough for further investigation.

A **Tuned Decision Tree** was also tested as a strong challenger model. It produced an **Accuracy of 0.6376**, **Precision of 0.1319**, **Recall of 0.9027**, **F1 Score of 0.2302**, and **ROC-AUC of 0.7963**. While its **F1 Score and ROC-AUC were slightly lower** than the Original Random Forest, it achieved the **highest Recall** among the models tested. This means it was the most aggressive in detecting fraudulent claims and could be considered a strong alternative if the business priority is to maximize fraud capture, even at the cost of more false positives.

The **Baseline Decision Tree** was also evaluated and achieved an **Accuracy of 0.9095**, **Precision of 0.2329**, **Recall of 0.2216**, **F1 Score of 0.2271**, and **ROC-AUC of 0.5875**. Although it appeared strong on Accuracy and Precision, its relatively low Recall indicated that it missed a significant portion of fraudulent claims, making it less suitable as the primary fraud screening model.

As part of the **Ethical AI and mitigation review**, the **Mitigated Random Forest** was assessed to observe how stricter thresholding or conservative governance settings affect performance. The mitigated version achieved **Accuracy of 0.9403**, **Precision of 1.0000**, **Recall of 0.0054**, **F1 Score of 0.0108**, and **ROC-AUC of 0.8167**. The confusion matrix showed **TN = 2899, FP = 0, FN = 184, TP = 1**, meaning the model flagged only **1 fraudulent claim out of 185 actual fraud cases** and correctly identified that one. While the precision was mathematically perfect, the model became **far too conservative** and failed to detect most fraudulent claims. This demonstrates that ethical safeguards and risk controls must be implemented carefully, because a model can become technically "safe" yet operationally ineffective.

The **Gradient Boosting** model also showed relatively good ranking ability but was not selected because it missed too many fraud cases in actual classification decisions. This made it less practical for real-world fraud screening.

Purpose of the Mitigated Model:

The mitigated Random Forest was not developed to replace the selected final model automatically. Instead, it was intentionally used as an ethical AI stress test to examine how the removal of potentially sensitive or proxy-risk variables affects predictive behavior, fairness, and model generalization. This distinction is important because ethical validation should inform governance decisions without automatically overriding the most operationally effective model.

Final Model Selection

Based on the full validation results, the **Original Random Forest** was selected as the **final predictive tool** because it delivered the **best overall balance** between:

- detecting a high proportion of fraudulent claims,
- maintaining acceptable screening quality,
- and achieving the **highest F1 Score** among the practically viable models.

Why Precision = 1.0000 is misleading in this case:

The mitigated model's perfect precision should not be interpreted as proof of superior fraud detection quality. The confusion matrix shows that the model predicted only 1 fraud case in the test set, and that single prediction happened to be correct. Because the model made almost no positive predictions, it achieved mathematically perfect precision while still failing to detect the overwhelming majority of actual fraud cases. This is why Recall and F1 Score remain critical for interpretation in imbalanced fraud problems.

Governance Lesson from the Mitigation Test:

The mitigation experiment demonstrates that removing potentially risky features may preserve ranking ability but can also create unintended operational consequences. In this case, the mitigated model became extremely conservative and showed strong overfitting, which means fairness-aware model development must be combined with threshold tuning, class balancing, and further validation rather than relying on feature removal alone.

Business Recommendation

The final model should be used as a **fraud screening and investigation triage tool**, not as an automatic claim denial engine. Claims flagged by the model should be routed for **human review, fraud investigation, and supporting evidence validation** before any final claims decision is made.

Key Validation Insight

This validation exercise also showed that:

- a model with **high Accuracy alone can still be weak for fraud detection**,
- **perfect Precision can be misleading** if Recall is extremely low,
- and the **best fraud model is the one that balances detection power with operational practicality**, rather than optimizing a single metric in isolation.